8 Cars are made in a factory for 24 hours every day.

In the factory a car is made every 209 seconds.

(a) Work out an estimate for the number of cars made in the factory in one year.
You must show how you get your answer.

365→400days;209→200seconds per car.

1 mark
Round two values

Seconds per day = 60×60×24 = 86400.

1 mark
Seconds in a day

Estimated seconds per year = 86400×400.

Cars ≈86400×400÷200 = 86400×2.

1 mark
Full yearly process

=172800 cars.

1 mark
172800

(4)

(b) Is your answer to part (a) an underestimate or an overestimate?
Give a reason for your answer.

Overestimate:more days(400) and fewer seconds per car(200)

both increase the estimated number of cars.

1 mark
Both changes raise count

(1)

(Total for Question 8 is 5 marks)